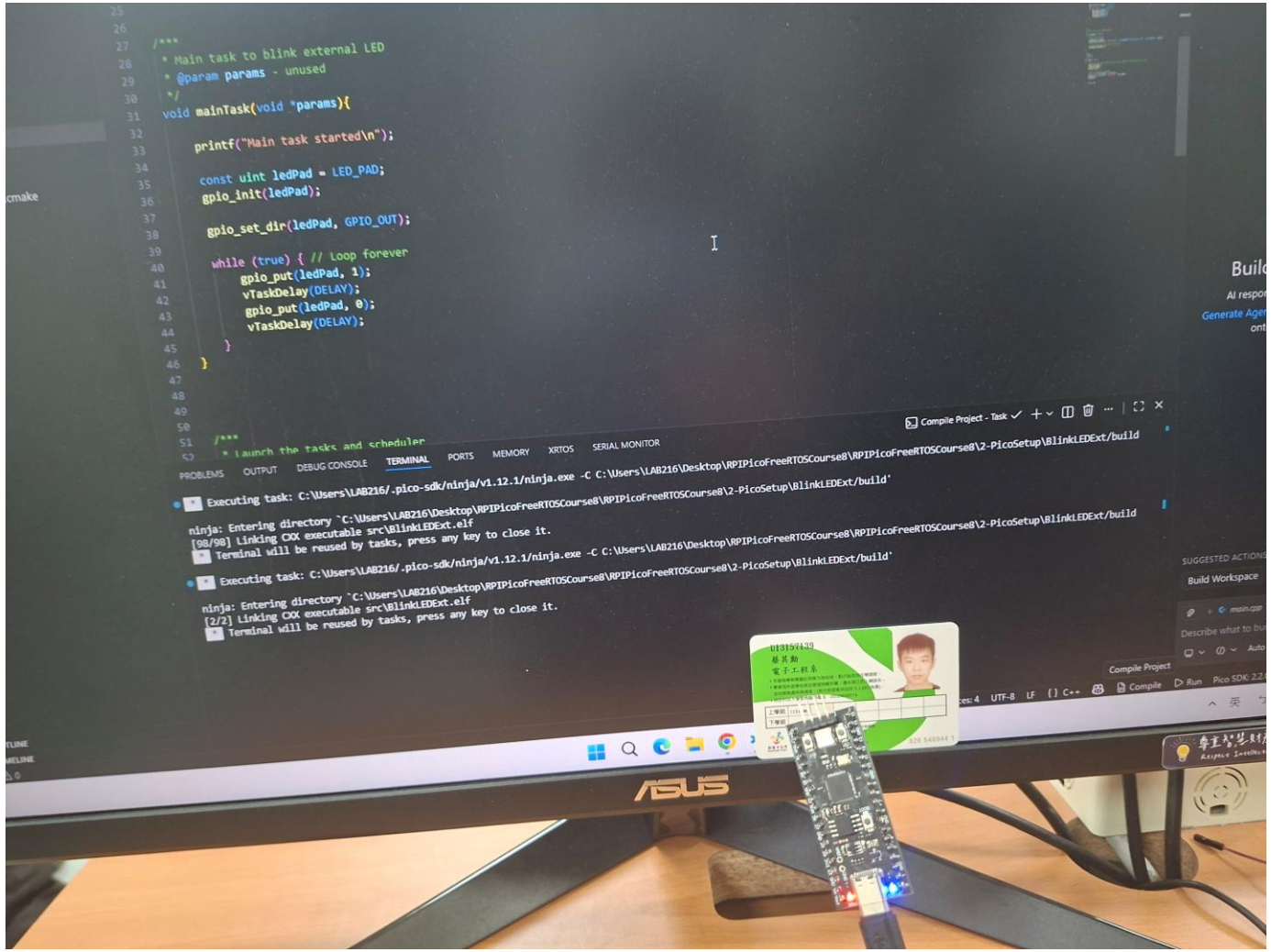
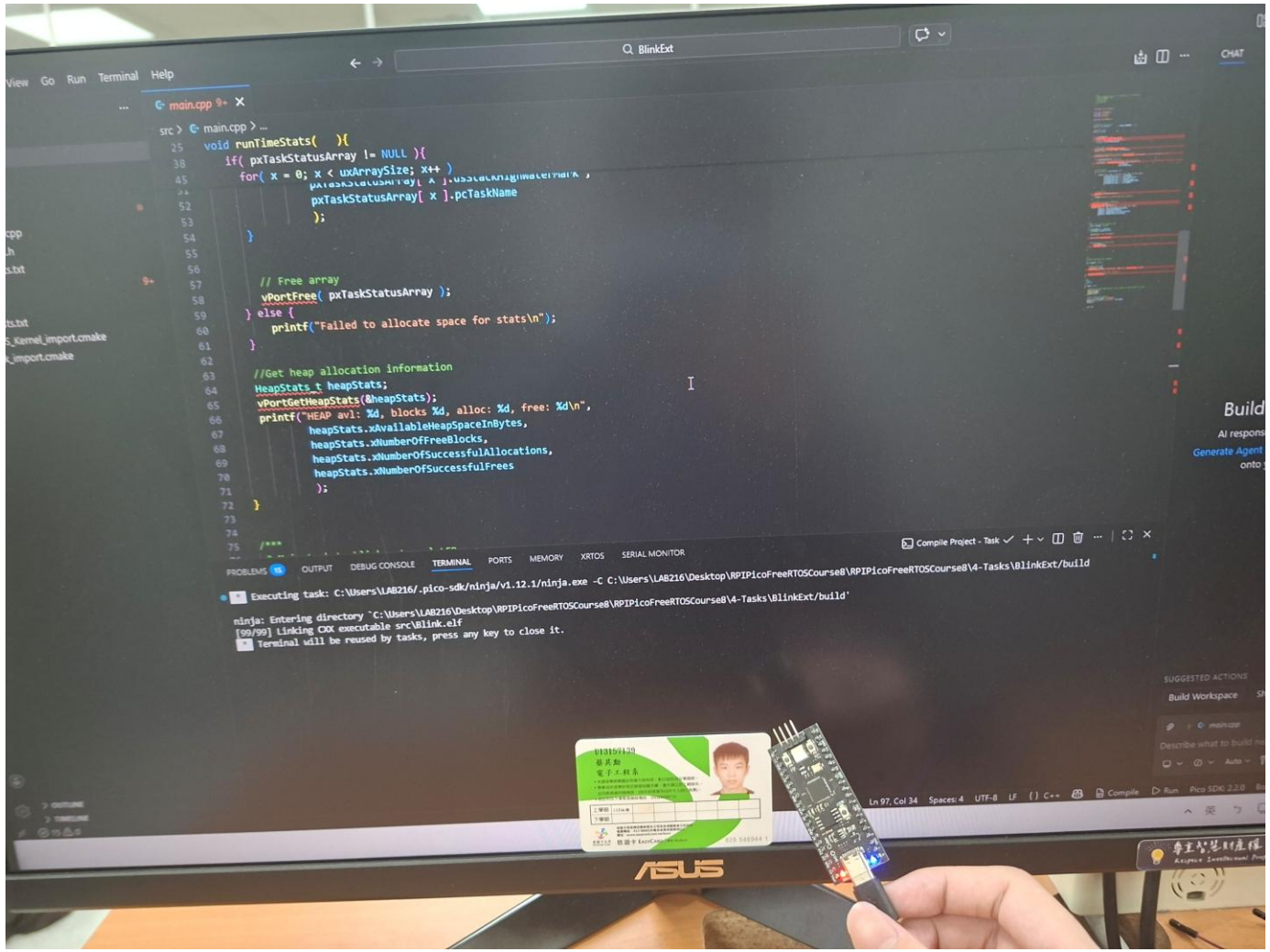
OS-3

U13157139

CH2


/***
* Main task to blink external LED
* @param params - unused
*/
void mainTask(void *params){
printf("Main task started\n");
const uint ledPad = LED_PAD;
gpio_init(ledPad);
gpio_set_dir(ledPad, GPIO_OUT);
while (true) { // Loop forever
gpio_put(ledPad, 1);
vTaskDelay(DELAY);
gpio_put(ledPad, 0);
vTaskDelay(DELAY);
}
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS MEMORY XRTOS SERIAL MONITOR
Compile Project - Task
Executing task: C:\Users\LAB216/.pico-sdk/ninja/v1.12.1/ninja.exe -C C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\2-PicoSetup\BlinkLEDExt/build'
ninja: Entering directory `C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\2-PicoSetup\BlinkLEDExt/build
[98/98] Linking CXX executable src\BlinkLEDExt.elf
Terminal will be reused by tasks, press any key to close it.
Executing task: C:\Users\LAB216/.pico-sdk/ninja/v1.12.1/ninja.exe -C C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\2-PicoSetup\BlinkLEDExt/build'
ninja: Entering directory `C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\2-PicoSetup\BlinkLEDExt/build
[2/2] Linking CXX executable src\BlinkLEDExt.elf
Terminal will be reused by tasks, press any key to close it.
Build Workspace
Compile Project
ASUS

CH4


BlinkExt
main.cpp
void runTimeStats( ){
if(pxTaskStatusArray != NULL){
for(x = 0; x < uxArraySize; x++)
pxTaskStatusArray[x].pcTaskName
// Free array
vPortFree(pxTaskStatusArray);
} else {
printf("Failed to allocate space for stats\n");
//Get heap allocation information
HeapStats_t heapStats;
vPortGetHeapStats(&heapStats);
printf("HEAP avl: %d, blocks %d, alloc: %d, free: %d\n",
heapStats.xAvailableHeapSpaceInBytes,
heapStats.xNumberOfFreeBlocks,
heapStats.xNumberOfSuccessfulAllocations,
heapStats.xNumberOfSuccessfulFrees
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS MEMORY XRTOS SERIAL MONITOR
Executing task: C:\Users\LAB216/.pico-sdk/ninja/v1.12.1/ninja.exe -C C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\4-Tasks\BlinkExt/build
ninja: Entering directory `C:\Users\LAB216\Desktop\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\4-Tasks\BlinkExt/build'
[99/99] Linking CXX executable src\Blink.elf
Terminal will be reused by tasks, press any key to close it.
ASUS

READ_ME

- CH2
- 使用 BlinkAgent 類別控制 LED 額外實作 runTimeStats()： 列出所有 FreeRTOS Task 狀態 顯示： Task 數量 Current Priority Base Priority Stack High Water Mark Task 名稱 顯示 Heap 使用情況 每 3 秒印出一次系統狀態 xTaskCreate() 使用 stack size = 500
- CH4
- 不使用 BlinkAgent 直接用 Pico SDK GPIO API 不印 Task 狀態 不印 Heap 資訊 使用 configMINIMAL_STACK_SIZE 單純 LED 每 500ms 閃爍